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REPORT



Interplay of heavy chain introns influences efficient transcript splicing and affects product quality of recombinant biotherapeutic antibodies from CHO cells

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ABSTRACT

Introns are included in genes encoding therapeutic proteins for their well-documented function of boosting expression. However, mis-splicing of introns in recombinant immunoglobulin (IgG) heavy chain (HC) transcripts can produce amino acid sequence product variants. These variants can affect product quality; therefore, purification process optimization may be needed to remove them, or if they cannot be removed, then in-depth characterization must be carried out to understand their effects on biological activity. In this study, HC transgene engineering approaches were investigated and were successful in significantly reducing the previously identified IgG HC splice variants to <0.5%. Subsequently, a comprehensive evaluation was conducted to understand the influence of the different introns in the HC genes on the expression of recombinant biotherapeutic antibodies. The data revealed an unexpected cooperation between specific introns for efficient splicing, where intron retention led to significant reductions in IgG expression of up to 75% for some intron combinations. Furthermore, it was shown that HC introns could be fully removed without significantly affecting productivity. This work paves the way for future biotherapeutic antibody transgene design with regard to inclusion of HC introns. By removing unnecessary introns, transgene mRNA transcript will no longer be mis-spliced, thereby eliminating HC splice variants and improving antibody product quality.

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Introduction

High-yielding recombinant protein expression is fundamental for biopharmaceutical production. Monoclonal antibodies (mAbs), which occupy a substantial fraction of the total biotherapeutic market, are predominantly produced in heterologous expression systems of mammalian origin such as Chinese hamster ovary (CHO) cells.¹ As product heterogeneity can arise between manufacturing batches,² regulatory guidelines require critical quality attributes (CQA) to be monitored.^{3,4} For mAbs, CQAs include the protein primary structure, variants of which can result in altered antibody structure that, in turn, can affect potency, pharmacokinetics, and product safety.^{5,6} Therefore, identifying and characterizing any sequence variants present is essential during product development.

The inclusion of introns in genes encoding therapeutic proteins is a strategy that is routinely used to improve the expression of these proteins.⁷ However, HC intron mis-splicing during expression of mAbs has led to premature stop codon-induced truncated protein and to the production of antibody isoforms with unwanted extensions, negatively affecting expression and product quality.^{8–10} Usually, only low levels of the mis-spliced transcript variants are produced compared to the correct full-length transcript and, therefore, the presence of the corresponding protein variants may not be identified until later stages of development, when the purified product undergoes detailed characterization. The presence of unwanted protein isoforms

with altered amino acid sequence or added domains could influence antibody–target interaction and potentially alter affinity for receptors and complement proteins and thereby affect potency.⁶ Thus, the presence of unwanted splice variants during antibody expression is not only a drain on the biosynthetic resources of the expression system but also in the subsequent efforts for their removal during downstream purification and assessment of patient safety.

Removing introns in their entirety has not been a preferred approach to eliminate the occurrence of splice variants because multiple lines of evidence indicate that the presence of an intron boosts transgene expression.^{11–13} Early observations of intron-mediated improvements in recombinant protein titer include the addition of an intron in the beta globin gene, which led to a 400-fold increase in protein expression.¹⁴ Similarly, insertion of generic heterologous introns between the promoter and gene of interest led to increases in expression for several transgenes in mice.^{11,15} There are several possible mechanisms of intron-mediated increases in transcript and protein expression; for example, the presence of introns can lead to enhanced transcription by influencing the promoter directionality and by harboring enhancer elements.^{16,17} Further to this, there is evidence of introns influencing the local chromatin structure and positioning of nucleosomes.^{18,19} In addition, there are multiple reports of splicing factors engaging in crosstalk with other RNA processing regulators

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participating in 3'-end processing, cleavage, and polyadenylation.²⁰ Therefore, the presence of an intron can often vouchsafe for subsequent faithful processing and export of the transcript to the cytosol.²¹

Examples of contaminating antibody splice variants have been previously reported. Spahr et al.⁸ identified a low level (~1–5%) C-terminal Fc extension impurity in multiple antibody molecules derived from stably transfected CHO cells. Subsequent fragment analysis revealed that this impurity resulted from the aberrant recognition of the C-terminal glycine codon as a 5' splice site donor, leading to an unexpected splice event with downstream plasmid sequence resulting in a longer Fc domain. Similarly, Harris et al.⁹ identified a 11 kDa Fc C-terminal extension sequence variant of the recombinant IgG1 which arose from an unexpected splicing event between a cryptic splice site donor in the 3' end of the HC gene and a splice acceptor site in the 5' end of the neighboring light chain (LC) constant domain. Most recently, Delmar et al.¹⁰ identified two low molecular weight IgG1 fragments resulting from intron mis-splicing, one caused by a cryptic splice acceptor site present in the CH2 domain resulting in expression of a truncated HC in the hinge region (Fab+Ala) and the other fragment derived from intron retention resulting in a variant containing translated intronic sequence (Fab +30aa). These studies highlight the need for further mechanistic understanding to prevent the occurrence of these intron mis-splicing events that give rise to unwanted antibody variants.

Our observations during mAb expression studies have consistently shown that the presence of introns in the heavy chain constant domain (HCCD) results in higher antibody titers compared to intron-less versions of the same expression plasmids for multiple mAbs, yet we know that using introns comes with the risk of generating splice variants. Here, we investigated several HC transgene engineering approaches to eliminate the Fab+Ala splice variant identified by Delmar et al.¹⁰ whilst retaining the HCCD introns. This was followed by a detailed investigation of the influence of individual introns in the HCCD on antibody titer and product quality for several different mAbs. Our studies have defined the contribution of individual introns on mAb expression and uncovered previously unknown cooperative mechanisms of splicing of HC introns. In addition, our results support the elimination of HCCD introns from antibody expression constructs which have previously led to mis-spliced aberrant mAb isoforms. This work provides a refined approach for HC transgene construct engineering to ensure the most efficient mAb expression and product quality.

Results

Intron 2 of the IgG HC constant domain is susceptible to mis-splicing resulting in the expression of aberrant protein species

Transgenes used for the expression of biotherapeutic mAbs may contain endogenous introns. There are three naturally occurring introns in the HCCD of IgG1, one between the CH1 and hinge (referred to here as HCCD-intron1), the second between the hinge and CH2 (referred to here as HCCD-intron2) and the third between the CH2 and CH3 domains

(referred to here as HCCD-intron3), as illustrated in Figure 1a. Previously, it was demonstrated that for mAb-A, a recombinant IgG1 with triple mutation (TM) modifications,²² expression from a 'gDNA' plasmid containing all three HCCD introns, resulted in three IgG1 variants caused by mis-splicing events.^{9,10} Two of these variants were directly caused by mis-splicing of HCCD-intron2. The Fab+Ala variant resulted from a cryptic splice acceptor site present in the CH2 domain which had a strong splice acceptor score as determined by a splice site detection tool.²³ The HCCD in mAb-A had been sequence optimized for expression in CHO cells, while retaining the introns and sequences flanking the intron-exon junctions as non-optimized in order to prevent any negative effect on splicing. However, the cryptic acceptor introduced by sequence optimization of the CH2 coding domain evidently affected splicing and resulted in expression of aberrant protein species.¹⁰ To investigate whether this cryptic splicing event was likely to be variable-domain independent and a potential reoccurring problem for molecules using the gDNA sequence-optimized IgG1, TM HCCD, expression plasmids were generated for two different mAbs: mAb-A (the molecule which is the focus in Harris et al.⁹ and Delmar et al.¹⁰ as a positive control for the Fab+Ala splice variant) and mAb-B, both with IgG, TM constant domains, and the gDNA sequence optimized arrangement in the HCCD. These plasmids were used to generate CHO stable pools. Sequence analysis of individual HC transcripts from these pools was performed by isolation of RNA followed by RT-PCR of the HC mRNA using primers designed to amplify the hinge-CH2 region. The resulting individual cDNA fragments were cloned by TOPO-TA cloning, and individual clones were analyzed using Sanger sequencing. Sequences of the correctly spliced cDNA and the Fab+Ala variant are shown in Supplementary Figure S1. This cDNA sequence analysis revealed that the Fab+Ala variant was present in both mAb-A and mAb-B expressing cells at a relatively similar level (8.8% and 6.5%, respectively, Table 1), indicating that the occurrence of this cryptic splice variant was independent of the two different antibody variable domain sequences.

Three transgene engineering approaches were explored to eliminate this cryptic splicing event. The first was the construction of a 'cDNA' sequence-optimized version of the HCCD omitting HCCD introns 1, 2 and 3 (cDNA-opt), but retaining the introns in the signal peptide and between the VH-CH1. The second involved construction of a non-sequence-optimized IgG1, TM HCCD sequence (gDNA) retaining all introns and lacking any high-scoring splice acceptors in close proximity to the true splice acceptor location at the boundary of the HCCD-intron2 and the CH2 exon. The third strategy was mutation of the high-scoring cryptic splice acceptor responsible for the Fab+Ala variant. A silent single base change in the cryptic splice acceptor *in silico* made it undetectable according to the splice site scoring tool,²³ but resulted in the appearance of a potential alternative acceptor site downstream at the next AG, which is the acceptor site consensus sequence. To combat this new site, two additional silent base changes were introduced to lower the splice acceptor score (gDNA-opt-mut). These constructs were used to generate stable CHO pools expressing mAb-A or mAb-B. The productivities of the pools were assessed by a fed-batch

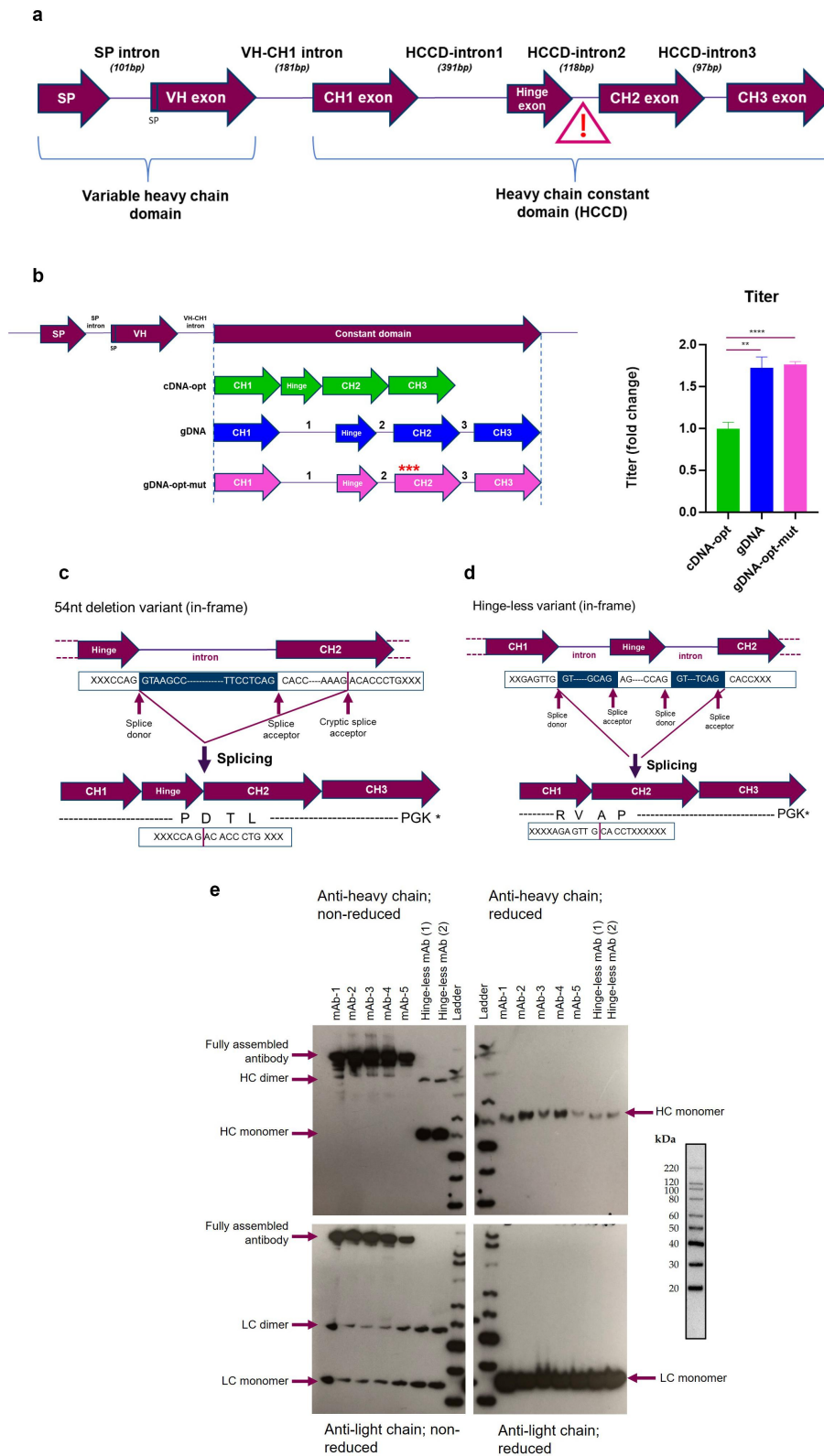


Figure 1. Heavy chain transgene engineering approaches to eliminate splice variants (a) a pictorial representation of the exon/intron arrangement in the HC expression cassette, the intron between the hinge and CH2 domain is highlighted as particularly susceptible to mis-splicing, SP = signal peptide (b) Depiction of the different HCCD intron arrangements generated in a single plasmid encoding both the HC and LC for mAb-B, the different HCCD formats were cDNA sequence optimized (cDNA-opt), gDNA non-sequence-optimized (gDNA) and gDNA sequence optimized with three mutations in the CH2 to lower the score of *in silico* predicted cryptic splice acceptors, illustrated by the red stars (gDNA-opt-mut). Plasmids were used to generate stable CHO pools which were subject to a fed-batch process, $n = 3$ pools for each construct, graph shows mean+SD. Terminal fed-batch titers represented as fold change compared to the cDNA sequence optimized plasmid, statistics analysis was performed using an unpaired t-test, * = $P < 0.05$, ** = $P < 0.005$, *** = $P < 0.005$, **** = $P < 0.0005$. (c) Molecular mechanism of the variant transcript produced by mis-splicing using a cryptic acceptor site resulting in deletion of the first 54 nucleotides of the CH2 domain. (d) Molecular mechanism of the hinge-less splice variant transcript, illustrated by the red stars (gDNA-opt-mut). Both variant transcripts resulted in in-frame shifts in coding sequence, * = Stop codon. (E) Non-reduced and reduced western blot analysis of mAb-B HC (top panels) and LC (lower panels) from transient expression supernatants, sampled at day 5 post transfection; mAb-1 to 5 are the five expression constructs in Figure 2a, present as positive controls, hinge-less mAb 1 and 2 are two transfection replicates of the same plasmid.

Table 1. Heavy chain transgene engineering summary.

Heavy chain format	Splice variant transcript present?		% Transcript occurrence of splice variant		Description
	mAb-A	mAb-B	mAb-A	mAb-B	
gDNA-opt	✓	✓	8.8	6.5	Original splice variant from Delmar et al, 2022 ¹⁰ (Fab+Ala)
cDNA-opt	×	×	NA	NA	No splice variants detected
gDNA-opt-mut	Not evaluated	✓	Not evaluated	1.1	Splice variant detected; 54 nucleotide deletion at 5' of CH2
gDNA	Not evaluated	✓	Not evaluated	0.44	Splice variant detected; hinge-less transcript

Description of the plasmid engineering approaches taken to eliminate the Fab+Ala cryptic splice variant using expression plasmids depicted in Figure 1b, and the outcome of each approach. % transcript occurrence detected by RT-PCR using primers spanning the hinge-CH2 region followed by TOPO cloning and colony PCR of initially 96 colonies per format, a further 672 colonies were analyzed for gDNA format due to the detection of the hinge-less splice variant occurring at a frequency of less than 1%. PCR products were analyzed using Sanger sequencing. Opt = sequence optimized for expression in CHO, mut = triple mutation to lower *in silico* predicted score of cryptic splice acceptors.

culture process, and cell pellets were analyzed for the presence of splice variants. As expected, the cDNA expression plasmid (cDNA-opt) eliminated the occurrence of the cryptic splice variant (Table 1) as it was missing all three of the endogenous HCCD introns, including HCCD-intron2 which was previously shown to contribute to the presence of splice variants. However, this plasmid showed a reduction in mAb-B titer of 1.7- and 1.8-fold compared to the gDNA-opt-mut and gDNA constructs, respectively (Figure 1b). Transcript sequencing analysis revealed that the gDNA-opt-mut construct, which had been mutated to eliminate the original cryptic splice site (responsible for the Fab+Ala variant) and to lower the score of a new splice acceptor site, surprisingly resulted in the low-level occurrence of a transcript with an in-frame deletion of the first 54 nucleotides of the CH2 (Supplementary Figure S1). This was driven by the new cryptic splice acceptor site detected *in silico* despite sequence engineering to reduce its splice acceptor score. If translated, this would result in an aberrant protein with a deletion of the first 18 amino acids of the CH2 (Figure 1c). This variant was detected at 1.1% transcript occurrence, compared to the original Fab+Ala variant at 6.5% (Table 1). A further unique splice variant omitting the hinge exon (0.44% occurrence) was detected arising from the gDNA non-optimized construct (Figure 1d, Supplementary Figure S1). To investigate the consequences of this hinge-less HC, a mAb-B expression plasmid omitting the hinge region of the HCCD was generated and assessed by transient transfection of CHO cells alongside mAb-B plasmids containing the hinge. Reduced and non-reduced western blotting of day-5 post-transfection supernatants detected the presence of secreted hinge-less HC, but this did not assemble with the LC as no fully formed antibody could be detected (Figure 1e). This hinge-less HC could be removed during downstream processing and is therefore considered a low-risk variant.

Unexpectedly, the gDNA non-sequence optimized HCCD (gDNA) plasmid resulted in similar mAb-B titers to sequence optimized gDNA-opt-mut when stable CHO pools were assessed in a fed-batch process (Figure 1b), therefore there was no expression benefit using the optimized format over the non-sequence optimized. Further investigations directly comparing the expression of sequence optimized with an equivalent non-optimized HCCD arrangement without splice mutations revealed that sequence optimization did not result in expression enhancement (Supplementary Figure S2).

Taken together, the cDNA sequence analysis revealed that the intron-less HCCD (cDNA-opt) was the only construct that successfully eliminated HCCD splice variants. However, the

removal of the HCCD introns also resulted in a significant reduction in titer compared to the intron-containing gDNA constructs (gDNA-opt-mut and gDNA), supporting the evidence in the literature that introns have a significant role in influencing expression.

IgG HCCD-Intron2 is dispensable, but HCCD-Intron1 is required to maintain high levels of intron-mediated expression enhancement

The contribution of each intron in the HCCD to expression enhancement was investigated as data in Figure 1b show that removal of all three endogenous IgG1 HCCD introns from the mAb-B (cDNA-opt) resulted in a significant reduction in titer in stable CHO pools. A panel of mAb-B HCCD non-sequence-optimized expression plasmids was constructed (Figure 2a), where each intron was systematically removed resulting in deletion of either HCCD-intron1 (gDNAΔ1), HCCD-intron2 (gDNAΔ2), or HCCD-intron3 (gDNAΔ3) in addition to deletion of all three introns simultaneously (cDNA). The HCCD gDNA non-optimized sequence was chosen as the basis for these constructs due to the higher mis-splicing occurrence of HCCD-intron2 in the gDNA optimized sequence compared to the gDNA non-optimized sequence (Table 1). CHO pools stably transfected with these constructs were compared in a fed-batch process (Figure 2b). Surprisingly, titers from all pools generated from the constructs with one deleted HCCD intron were comparable to the mAb-B gDNA control pool, regardless of which HCCD intron had been removed. Cell culture parameters monitored during the fed-batch process (Figure 2c–e) demonstrated that all pools performed similarly in terms of growth and viability and that specific productivity mirrored the titers (Figure 2f).

As HCCD-intron2 had been shown to drive mis-splicing (Figure 1) and its removal had no significant effect on titer (Figure 2b), it was desirable to eliminate this intron from the HCCD. Further evaluations focused on mAb-B expression constructs, with the HCCD containing HCCD-intron1 (cDNA +1) or HCCD-intron3 (cDNA +3) alone (Figure 3a). The mAb-B construct gDNAΔ2, containing HCCD introns 1 and 3, but not intron 2, served as a positive control as titers from these stable pools were similar to those generated using the gDNA construct, with all three HCCD introns present (Figure 2b). Stable CHO pools were generated for this panel of constructs and assessed in a fed-batch process for titer (Figure 3b), viable cell number, viability, and specific

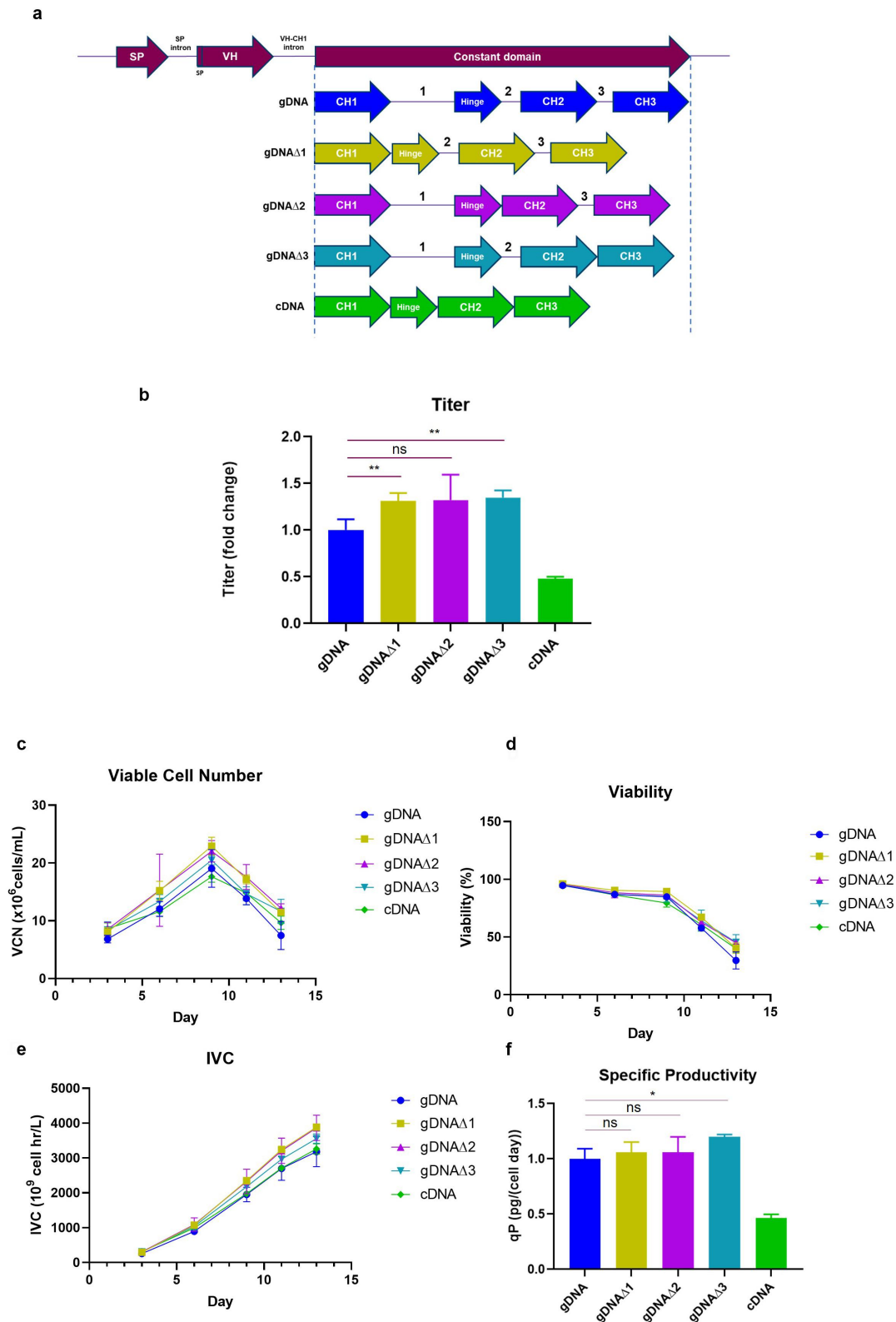


Figure 2. Removal of individual introns in the HCCD does not affect titer. (a) Depiction of the different HCCD intron arrangements generated in a plasmid expressing mAb-B showing systematic removal of individual introns. Expression plasmids were evaluated using a stable CHO pool fed-batch process (b) Terminal fed-batch titers represented as fold change compared to gDNA (c) Average viable cell number (VCN) ($\times 10^6$ /mL) (d) Cell viability (%) (e) Integral of viable cells (IVC) (10^9 cell hr/L) (f) Cell productivity (qP) (pg/(cell day)) represented as fold change compared to gDNA. All graphs show the mean $\pm$ SD, $n = 3$ in all cases, statistical analysis was performed using an unpaired t-test, ns= not significant, * = $P < 0.05$, ** = $P < 0.005$.

productivity (Supplementary Figure S3). Titers from cDNA +1 mAb-B expressing pools containing HCCD-intron1 alone, were similar to the gDNAΔ2 control pools. However, the titers

from cDNA +3 pools containing HCCD-intron3 alone were 2-fold lower than cDNA +1 pools and similar to cDNA pools, missing all three introns (Figure 3b). These data indicate that

HCCD-intron3 has minimal effects on expression, whereas HCCD-intron1 has a significant positive impact, boosting expression by 2.7-fold compared to cDNA pools.

HCCD-intron1 enhances expression by promoting efficient splicing of the VH-CH1 intron in a position-independent manner.

To further explore the influence of HCCD-intron1 on IgG1 expression, investigations were widened to consider any effects of the introns present upstream of the HCCD (Figure 1), namely the introns within the signal peptide and between the VH and CH1 domains. Three mAb-B HCs were generated containing the signal peptide intron alone (HC+SP), HCCD-intron1 alone (HC + 1, with no signal peptide intron or VH-CH1 intron) and a completely intron-less HC (HCAI), as illustrated in Figure 3c. These HC expression cassettes were assessed in stable CHO pools in comparison with mAb-B gDNA, cDNA +1 and cDNA expressing pools for productivity (Figure 3d and Supplementary Figure S3).

The titer data show that there is no significant difference between the gDNA construct containing all five introns (gDNA) and the HC-intron-less construct (HCAI) (Figure 3d), implying that these HC introns have minimal influence on expression and can be removed without significantly affecting the product titer. There is, however, a general upward trend in titers for the gDNA and the cDNA +1 constructs compared to the HC-intron-less construct (HCAI) even though it is not statistically significant. The titer data also show that the presence of either the signal peptide intron (HC+SP) alone or HCCD-intron 1 alone (HC + 1) in the HC cassette provides no increase in titer compared to the HC intron-less construct (HCAI). Surprisingly, mAb-B titers were significantly increased in pools transfected with the HC +SP construct, containing the signal peptide intron alone compared to cDNA containing both the signal peptide and VH-CH1 introns. These data indicate that the VH-CH1 intron has an inhibitory effect on expression if HCCD-Intron1 is not also present, as the addition of HCCD-Intron1 into the cDNA construct (cDNA +1) that contains the VH-CH1 intron significantly increases titer.

The effect of adding HCCD-intron1 back into the cDNA format was validated by evaluating additional three molecules alongside mAb-B. These were the previously used mAb-A (the focus in Harris et al.⁹ and Delmar et al.,¹⁰ and Figure 1), mAb-C and mAb-D, each containing unique variable LC and variable HC sequences coupled with three different HCCD arrangements, namely gDNA and cDNA +1 (containing HCCD-intron1 alone) as positive controls and cDNA (Figure 4a). The LCs for mAb-A and mAb-D were lambda, whereas mAb-B and mAb-C contained kappa LCs. CHO stable pools expressing these constructs were assessed for growth and productivity in a fed-batch process (Figure 4b and Supplementary Figure S4). The data demonstrate that the presence of HCCD-intron1 in the constant domain resulted in a 2.2 to 3.9-fold increase in titer (mAb-A 2.2-fold; mAb-B 3.3-fold; mAb-C 2.4-fold; mAb-D 3.9-fold) compared to the corresponding cDNA construct

depending on the molecule being expressed and that this uplift is independent of both HC variable domain and LC sequences.

As data indicated that the VH-CH1 intron was having an inhibitory effect on expression (Figure 3d), we investigated the splicing status of the VH-CH1 intron. cDNA was generated from the gDNA, cDNA and the cDNA +1 stable pools for all four molecules (mAb-A, mAb-B, mAb-C, and mAb-D, Figure 4a) and PCR was performed across the HC introns to assess the presence of introns. Firstly, it was established that efficient splicing of all introns occurred in the gDNA arrangement, as there was no evidence of introns present (Supplementary Figure S5). However, PCR using primers that were specifically annealed to either side of the VH-CH1 intron clearly showed that the cDNA arrangement for each molecule (i.e., containing no HCCD introns) had two bands present corresponding to intron-retained and intron-spliced products, whereas the gDNA or cDNA +1 constructs contained barely detectable intron-retained mRNA species (Figure 4c). It is likely that the faint bands present in the gDNA and cDNA +1 constructs correspond to pre-processed mRNA as a consequence of the total cellular cDNA synthesis method used, but clearly the intron-retained species is markedly increased in the cDNA construct. This directly correlated with the pool titers (Figure 4b), suggesting that the presence of HCCD-intron1 downstream of the VH-CH1 intron drives correct splicing of the VH-CH1 intron and that without HCCD-intron1, this intron is less efficiently spliced. If the VH-CH1 intron is retained, the predicted transcribed mRNA would contain a premature stop codon within the intron sequence, and therefore would presumably be targeted for non-sense-mediated decay, thereby decreasing the amount of correct mRNA available for translation and thus affecting pool titers. Sequence analysis of the gel-purified PCR products for mAb-D was confirmed as spliced or unspliced species (Supplementary Figure S6).

Data in this study show that the presence of HCCD-intron1 promotes efficient splicing of the VH-CH1 intron as in its absence the VH-CH1 intron is retained to a significant level (Figure 4c). To further understand the importance of HCCD-intron1 for this cooperative splicing and any positional effects, a series of mAb-B HC constructs were designed where HCCD-intron1 and HCCD-intron3 were inserted in different positions in the HCCD (Figure 5a). Inserting HCCD-intron3 to the HCCD-intron1 position, between CH1 and hinge of the HCCD (cDNA +3in1) caused a reduction in the predicted splice donor site score due to the surrounding sequence; therefore, in order to ensure an efficient donor site, the nucleotide sequence of HCCD-intron3 was modified to increase the strength of the 5' splice donor site score by making a 1-nucleotide silent change (cDNA +3in1-SM). These constructs were evaluated in stable CHO pools in a fed-batch process, in comparison with gDNA, cDNA +1, cDNA +3, and cDNA constructs (Figure 5b and Supplementary Figure S7). Titer data demonstrated that relocating HCCD-intron1 to the HCCD-intron3 position, between CH2 and CH3 of the HCCD (cDNA +1in3), resulted in similar titers to pools expressing the gDNA and cDNA +1 HC constructs. However, HCCD-intron3 inserted at the HCCD-intron1 position, between the CH1 and hinge (cDNA +3in1 and cDNA +3in1-SM), did not have

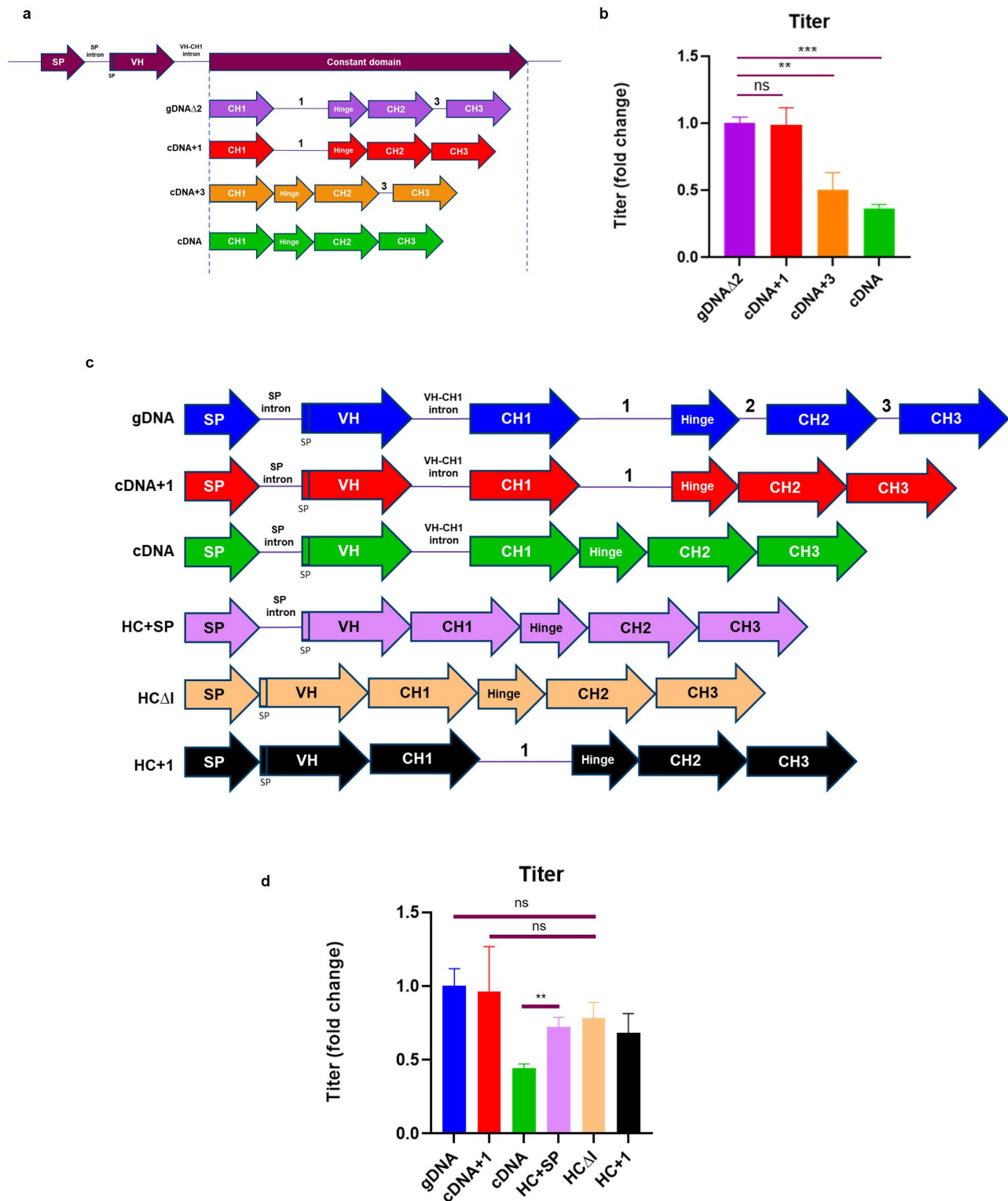


Figure 3. The VH-CH1 intron is inhibitory to titer unless HCCD-Intron1 is present (a) Depiction of the different HCCD intron arrangements generated in a plasmid expressing mAb-B. Expression plasmids were evaluated using a stable CHO pool fed-batch process. (b) Terminal fed-batch titers represented as fold change compared to gDNAΔ2. (c) Depiction of the different HCCD intron arrangements generated in plasmids expressing mAb-B. Expression plasmids were evaluated using a stable CHO pool fed-batch process (d) Terminal fed-batch titers represented as fold change compared to gDNA. The graphs show the mean + SD, $n = 3$ in all cases, statistics determined using an unpaired t-test, ** = $P < 0.005$, ns = not significant. (A). Depiction of the different HCCD intron arrangements generated in a plasmid expressing mAb-B where either intron2 and 3 have been removed simultaneously (cDNA+1) or intron1 and 2 have been removed simultaneously (cDNA+3), cDNA (containing no introns in the HCCD) and gDNA (containing introns 1 and 3 but without intron2 (gDNAΔ2) are present as controls. (b). A bar graph showing terminal fed-batch titers represented as a fold change compared to gDNA without intron2. The data show that there is no significant decrease in titer when introns 2 and 3 are removed simultaneously, but there is a significant decrease of approximately 50% when introns 1 and 2 are removed simultaneously, comparable to cDNA titers. (c). Depiction of the different HCCD intron arrangements generated in a plasmid expressing mAb-B including gDNA (containing all five introns), cDNA+1 (containing intron 1 only in the constant domain and the SP and VH-CH1 intron), cDNA (containing the SP and VH-CH1 intron only), HC+SP (containing the SP intron only), HCΔI (the HC containing no introns) and HC+1 (containing Intron1 only in the constant domain). (d). A bar graph showing terminal fed-batch titers represented as fold change compared to gDNA. The data show a significant increase in titer for HC+SP (the HC containing the signal peptide intron only) compared to cDNA (containing no introns in the constant domain) and no significant difference between gDNA (containing all five introns) or cDNA+1 (containing intron 1 only in the constant domain and the SP and VH-CH1 intron) and HCΔI (the HC containing no introns).

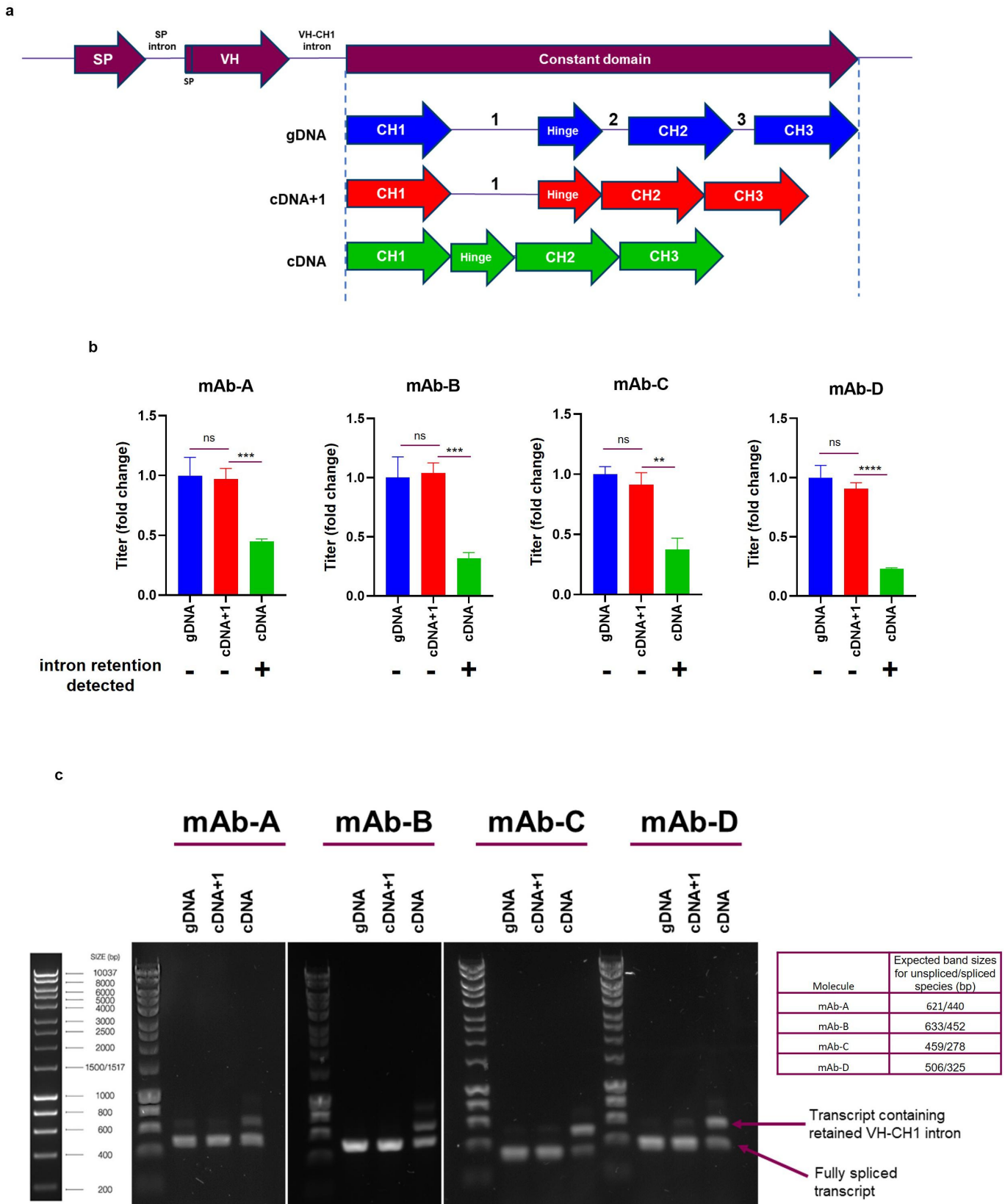


Figure 4. VH-CH1 intron is retained in the absence of HCCD-intron1 (a) Depiction of the different HCCD intron arrangements evaluated in plasmids expressing four different mAbs; mAb-A, mAb-B, mAb-C and mAb-D. Stable CHO pool were generated and evaluated using a fed-batch process. (b) Terminal fed-batch titers represented as fold change compared to gDNA for each molecule. The graphs show the mean + SD, $n = 3$ in all cases, statistical analysis was performed using an unpaired t-test, ** = $P < 0.005$, *** = $P < 0.0005$, **** = $P < 0.00005$, ns = not significant. Intron retention data from (c) is summarized beneath each construct tested, + = high level intron retention detected, - = little/no intron retention detected. (c) Evaluation of cDNA in a PCR based VH-CH1 intron retention assay using primers designed to anneal to each unique IgG variable HC and a common reverse primer in the CH1 constant region. PCR products were resolved and visualized on a 1% agarose gel. Molecular marker is Bioline HyperLadder 1kb. Table describes expected band sizes for spliced or unspliced species.

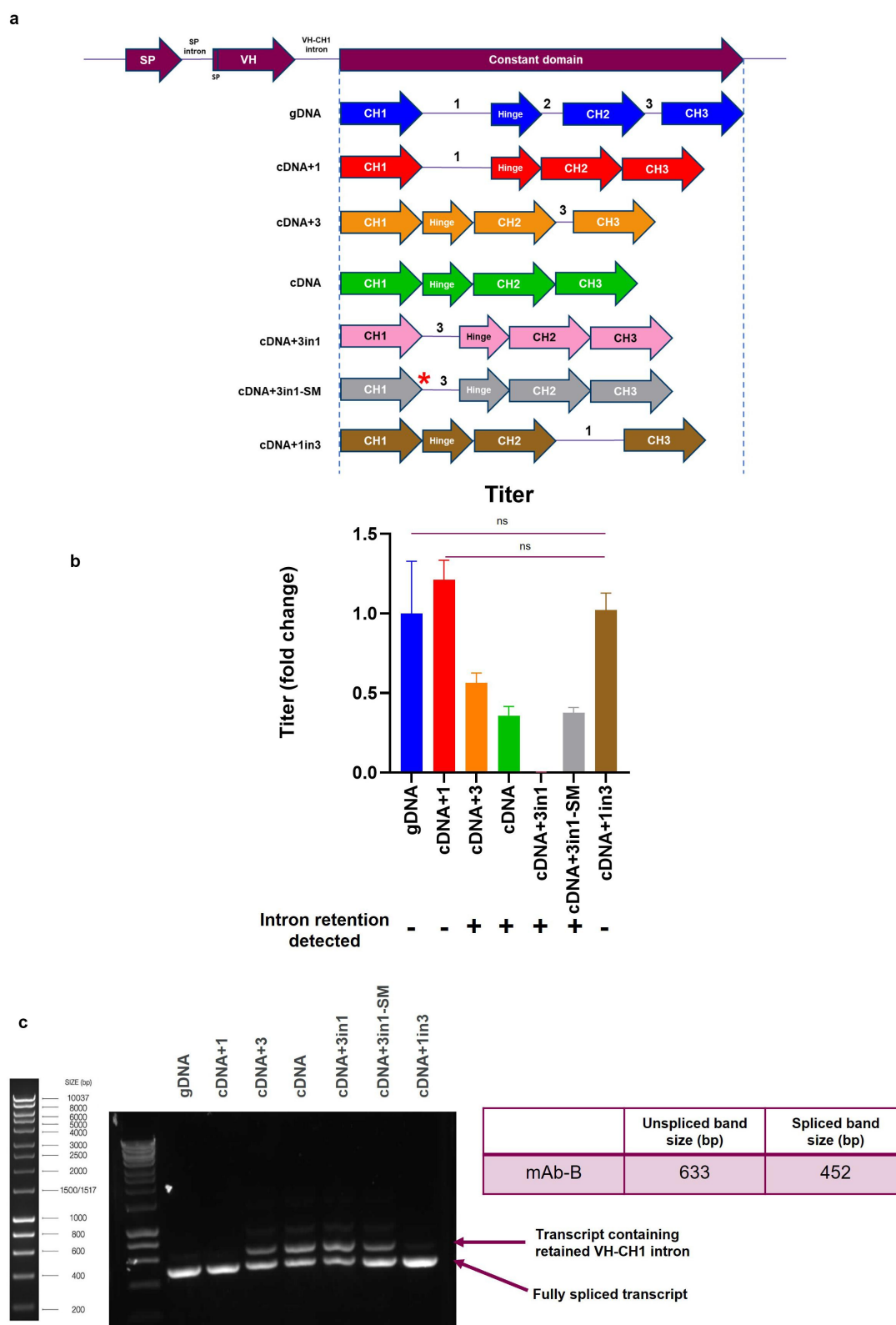


Figure 5. HCCD-intron1 can alleviate VH-CH1 intron retention when moved within the HCCD (A) Depiction of the different HC intron arrangements used in plasmids expressing mAb-B. Expression plasmids were evaluated using a stable CHO pool fed-batch process. SM= sequence modified by 1-nucleotide to increase splice donor strength (b) Fed-batch titers represented as fold change compared to gDNA. The mean + SD is shown, $n = 3$ in all cases, statistical analysis was determined using an unpaired t-test, ns=not significant. Intron retention data from (c) is summarized beneath each construct tested, + = high level intron retention detection, - = little/no intron retention detected. (c) Evaluation of the expression plasmids in a VH-CH1 intron retention assay. Total RNA was extracted from the CHO pool cells, followed by cDNA synthesis. PCR was carried out with primers designed to anneal to each unique variable HC and a common reverse primer in the constant region. PCR products were resolved and visualized on a 1% agarose gel. Molecular marker is Bioline HyperLadder 1kb. Table describes expected band sizes for spliced or unspliced species.

the same effect, resulting in lower production pools. Splicing analysis of these constructs (Figure 5c) revealed the presence of the VH-CH1 intron in the absence of HCCD-intron1, indicating that HCCD-intron1 has an important role in alleviating the VH-CH1 intron retention regardless of its position in the HCCD. However, unspliced species were detected in the presence of HCCD-intron3 demonstrating that HCCD-intron3 cannot alleviate the VH-CH1 intron retention. These data indicate that it is not merely the presence of any intron downstream of the VH-CH1 intron that promotes its efficient splicing, rather there is a specific role involving HCCD-intron1 in driving the efficient splicing of this upstream intron.

Discussion

The inclusion of introns in expression constructs for therapeutic protein production is a strategy often used to improve the expression of the transgenes⁷ and to enable rapid molecular cloning of different domains such as mAb variable domains.²⁴ However, the presence of introns in transcripts encoding recombinant proteins can be problematic if they are not correctly removed by the splicing machinery or if a cryptic splice site is used. Usually, only low levels of these transcript variants are produced compared to the correctly processed transcript. However, the corresponding contaminating product variants can be difficult to remove by purification and can cause potential safety concerns in terms of their impact on potency, pharmacokinetics, and immunogenicity. Therefore, it is important to minimize the occurrence of these transcript variants to ensure the high quality of therapeutic protein products.

In therapeutic mAbs, the intron between the hinge and CH2 of the HC gene has been shown to be particularly susceptible to mis-splicing, producing contaminating splice variants.¹⁰ Our initial approaches to combat these variants focused on preserving the use of intronic sequences in the HC cassette due to their accepted role in the enhancement of titer. These efforts included mutating non-canonical high scoring splice acceptor sites and using a non-sequence-optimized HCCD. However, neither approach fully eliminated splice variants, although using the non-sequence optimized HCCD significantly decreased the occurrence of transcript splice variants from 6.5% to 0.4%. The splicing reaction responsible for removing intronic sequences from transcripts requires accurate cleavage at the conserved splice site sequences found at the 5' and 3' ends of introns. These consensus sequences are known to be critical along with the branch point, which is located 18 to 40 nucleotides upstream from the 3' end of an intron and plays an essential role in the splicing reaction.²⁵ The 3' splice site (3'ss) is usually the first 'AG' downstream from the branch point site.²⁶ One possible reason why mutating the cryptic splice sites was unsuccessful in this study, is that even though these sites were located further downstream of the true 3'ss at the intron-CH2 boundary, they followed the pattern that if the 'AG' of the cryptic site was mutated, then a new cryptic splice acceptor appeared at the next 'AG' downstream. After introducing three silent mutations into this region, there remained a predicted cryptic splice acceptor site, albeit with a significantly lower score than the original cryptic site, but this

was evidently still being selected over the canonical 3'ss at a low level, resulting in a variant transcript.

Using a non-sequence-optimized version of the HCCD in our study resulted in low levels of a hinge-less HC antibody transcript. It is well known that smaller exons pose a problem for the splice machinery, potentially because the factors that bind the 3' and 5' splice sites flanking the exon are sterically hindered from binding simultaneously.²⁷ It is possible, therefore, that the hinge exon, being 45 nucleotides, falls under this category and is vulnerable to low-level exon skipping due to its small size. We went on to demonstrate that hinge-less HC transcript was successfully translated and secreted from CHO cells, but the protein product did not assemble with LC to form an intact antibody. The most likely reason for this is that the hinge plays an important role in the IgG1 structure by forming four inter-chain disulfide bonds; two between the two HCs and two between the HC and LC of each Fab arm,^{28,29} therefore in the absence of the hinge these disulfide bonds cannot form. Although it is unusual for HC to be secreted from cells without first pairing with LC, it has been previously observed.³⁰ However, it is also possible that the presence of this hinge-less HC in the cell culture medium might be a consequence of cell lysis, releasing otherwise retained intracellular protein. Although this particular splice variant was deemed low risk of being a contaminating species in the final biotherapeutic mAb product because it would be removed during downstream purification processes, the cells waste biosynthetic capacity producing this aberrant HC.

The identification of multiple splice variants prompted our thorough investigation into the requirement of introns in the HC expression cassette. Our results show that there is an interesting interdependency of splicing between the intron flanked by the VH and CH1 domains and HCCD-intron1, wherein if HCCD-intron1 was removed, the VH-CH1 intron splicing was severely affected, resulting in a dramatic decrease in antibody expression of up to 75%. We also found that removal of HCCD-intron1, while HCCD-intron2 and HCCD-intron3 were present did not decrease the antibody titers, yet HCCD-intron3 alone resulted in significant VH-CH1 intron retention and a decrease in titer comparable to cDNA (with no introns in the HCCD), further underscoring the interdependency of splicing between these introns and that HCCD-intron2 is also highly likely to influence efficient VH-CH1 intron splicing. However, due to its tendency to generate unwanted splice variants, inclusion of HCCD-intron2 in IgG1 expression plasmids is not recommended. Intron interdependency for splicing has been reported previously for human transcripts where some introns require neighboring splicing events for correct processing.³¹ Kim et al.³¹ also defined a category of introns that are only able to splice after neighboring splicing events have occurred and found that these dependencies appeared to be used by the cell to delay intron removal and enable alternative splicing pathways. With regard to the IgG1 HCCD introns, HCCD-intron1 and HCCD-intron2 may increase the splicing efficiency of the VH-CH1 intron in a cooperative manner, whereas HCCD-intron3 does not possess this quality. We found that HCCD-intron1 could also be moved approximately 400 bp downstream (into the location of HCCD-intron3, between the CH2 and CH3 domains) and

still relieve this VH-CH1 intron retention, whereas HCCD-intron3 could not promote efficient VH-CH1 intron splicing regardless of location within the HCCD. One possible explanation for this is that HCCD-intron3 has a shorter polypyrimidine tract compared to HCCD-intron1 and HCCD-intron2, which is known to be recognized by U2AF65 among other splicing factors.³² U2AF65 interacts with splicing factor 1 (SF1) and U2AF35, which recognize the branch point signal and the 3'ss, respectively.³³ Together the U2AF65-SF1-U2AF35 complex is responsible for aiding the stable assembly of small nuclear ribonucleoproteins (snRNPs) for accurate juxtaposition of splice sites and the subsequent catalysis. Therefore, HCCD-intron1 and HCCD-intron2 could potentially fulfill this role more effectively than HCCD-intron3.

A possible reason why the VH-CH1 intron is so heavily dependent on HCCD-intron1 for efficient splicing is that at the endogenous immunoglobulin HC locus, the intronic sequence between the VH and the constant region has an essential role in class switch recombination (CSR) allowing mature B cells to switch from IgM to IgG production.³⁴ Speculatively, it is possible that this sequence is optimized to enable its essential role in isotype class switching and therefore compromised for its function as an intron resulting in it relying more heavily on the presence of a powerful downstream intron for efficient splicing. The degree of VH-CH1 intron retention is also likely influenced by the VH sequence, suggesting that the exonic sequence influences splicing. This is evidenced by the 2- to 4-fold difference in the expression that HCCD-intron1 makes for four different VH sequences (Figure 4b). For example, insertion of the HCCD-intron1 sequence into the cDNA plasmid containing the variable domains for mAb-D resulted in a 75% increase in titer compared to a 54% increase for mAb-A. It would be interesting to investigate this further to elucidate the mechanism behind this observation. Similar evidence of an exonic sequence influencing intron splicing was noted for mouse IgM, wherein a specific enhancer sequence within the 5' portion of exon M2 was necessary for splicing of an upstream intron between exons M1 and M2.³⁵ Furthermore, the variable domains of the IgG molecules tested in our study were sequence-optimized for expression in CHO, and this optimization of the exonic sequences could have altered exon splice enhancing motifs, thereby affecting the splicing efficiency of the downstream VH-CH1 intron.³⁶

We have demonstrated here that efficient splicing occurs in the presence of all five endogenous introns in the HC (Supplementary Figure S5). It is only through manipulating the natural arrangement of introns to create biotherapeutic expression plasmids that problems with intron retention occur. In this instance, these introns are splicing interdependently and therefore need to be used in a specific arrangement to promote splicing fidelity. If used non-optimally, the VH-CH1 intron can have a significant inhibitory effect on IgG expression.

Interestingly, sequence optimization failed to improve the expression from two of the best-performing plasmids (gDNAΔ2 and cDNA +1). This is likely due to the highly optimized arrangement of other expression elements in these plasmids. It would be interesting to investigate if sequence optimization improved the expression of a completely intron-less-HC cassette as our study has revealed that introns

in the HC have minimal influence on expression and can be removed without significantly affecting antibody titers. This was surprising due to the well-documented role of introns in enhancing gene expression. A possible explanation of the benefits seen by including introns is that these are context specific and are brought about by crosstalk of splicing with transcription, mRNA decay, and transcript stability. In the context of biotherapeutic mAb expression plasmids, because a very strong heterologous promoter is driving HC expression and typically it is the HC expression that dictates mAb expression levels,³⁷ the promoter's activity likely supersedes everything else. It is possible that the interplay between the endogenous HC promoter and introns would be different with introns playing a more significant role in HC expression.

Our study demonstrates, for the first time, that there is cooperative splicing between specific introns within the HC cassette of biotherapeutic expression plasmids and that these widely accepted expression-enhancing elements are not acting as expected in this context, but quite the opposite, with the presence of the VH-CH1 intron inhibiting expression by up to 75% if not used in the optimal intron arrangement. This work has played a key role in advancing our understanding of the interplay between HC introns and transgene expression.

In conclusion, by removing introns from within the HC transgene, the transcribed mRNA can no longer be mis-spliced, thereby eliminating HC splice variants as contaminating species and ensuring the high quality of future biotherapeutic IgGs. This mitigates the need for time-consuming investigations to uncover the origins of splice variants, their effect on potency, pharmacokinetics, and safety of the product, and the process optimization to successfully remove them, ensuring consistency of production batches. These learnings from our study will be applied to, and inform, the future direction of gene construct design for optimal expression and product quality of our clinical biopharmaceuticals.

Materials and methods

IgG expression plasmids

Restriction digestion and ligation methods were used to generate HC expression constructs with specific intron arrangements in the antibody HC cassette (synthesized by GeneArt, ThermoFisher Scientific, Germany). This was followed by Gateway cloning to generate a single plasmid that encoded both the HC and LC in addition to a glutamine synthetase selectable marker. Variable domain sequences were sequence optimized for expression in CHO and synthesized by GeneArt (ThermoFisher Scientific, Germany). Unless otherwise stated, all plasmids used an IgG1 with the TM format as described by Oganessian et al.,²² where three amino acids in the CH2 domain differ from the IgG1 wild-type sequence for reduced IgG1 effector function.

Cell lines and culture conditions

An AstraZeneca CHO host, a derivative of CHO-K1, was maintained in CD CHO medium (Life Technologies, Cat# 10743029) supplemented with 6 mM L-glutamine (Life

Technologies, Cat# 25030–081). Stably transfected CHO cells were grown in an AstraZeneca production medium supplemented with methionine sulfoximine (MSX; Sigma – Aldrich, Cat# M5379). The cultures were grown in a humidified incubator at 36.5°C, 6% CO₂ with agitation at 140 rpm as required. For fed-batch culture, cells were cultured in AstraZeneca production medium as above without MSX over 11–14 days. The medium was supplemented with bolus additions of an AstraZeneca nutrient feed over the course of the culture period. Glucose and lactate were monitored throughout the fed-batch process using a YSI (2900D, YSI Inc). Cell count and viability analysis was performed using a Vi-CELL XR cell viability analyzer (Beckman Coulter, USA). Cell culture medium was clarified by centrifugation, and mAb titers were quantified by protein-A HPLC affinity chromatography on an Agilent 1260 Infinity series (Agilent Technologies, CA) by comparing the peak size from each sample with a calibration curve.

Site directed mutagenesis

Single or multiple nucleotide changes were made to gene sequences using either a QuikChange Lightning kit or a QuikChange multi-site directed mutagenesis kit following the manufacturer's instructions (Agilent, Cat# 210519 and 210,516, respectively).

Generation of IgG expressing stable pools

Stable CHO pools expressing IgGs were generated by transfecting CHO cells, with plasmids encoding the mAb of interest using an Amaxa nucleofactor and reagents (Lonza, Cat# VCA-1003). The transfected cells were selected and maintained in CD CHO medium in the presence of MSX. Pools of cells were expanded and evaluated for the production of IgG in a shaking fed-batch process.

Splice variant analysis

Sequence analysis of individual HC transcripts from CHO pools was performed by isolation of RNA using an RNAeasy mini kit (Qiagen, Cat# 74104), followed by RT-PCR of the HC mRNA using a Transcriptor one-step RT-PCR kit (Roche, Cat# 04655877001) and primers spanning the CH1- hinge-CH2 region. The resulting individual cDNA fragments were cloned into a plasmid using a TOPO-TA cloning plasmid kit and used to transform *E. coli*, according to manufacturer's instructions (Invitrogen, Cat# 450030). The resulting bacterial colonies were analyzed using Sanger sequencing.

Intron retention assay

Total RNA was extracted from cells using an RNAeasy mini plus kit (Qiagen, Cat# 74134). Complementary DNA was generated using SuperScript III First-Strand Synthesis SuperMix (Invitrogen, Cat# 18080–400). PCR was carried out with primers designed for each unique IgG variable HC and a common reverse primer in the constant region using DreamTaq Green PCR Master Mix (2X) (Thermo Scientific,

Cat# K1081) following the manufacturer's instructions. PCR products were resolved and visualized on a 1% agarose gel. Where appropriate, product bands were excised, purified using the QiaQuick gel extraction kit (Qiagen, Cat#28704) and analyzed using Sanger sequencing

Transient transfection and western blotting

For transient transfection, CHO cells were transfected with IgG expression plasmids using an Amaxa nucleofactor and reagents (Lonza, Cat# VCA-1003), recovered into CD CHO medium (Life Technologies, Cat# 10743029) in 24-well plates and incubated at 210 rpm, in a humidified incubator at 36.5°C, 6% CO₂. Culture supernatant was collected day 5 post transfection, diluted 1:10 before the addition of 4× loading buffer (Invitrogen, Cat# NP0007), β-mercaptoethanol (Millipore, Cat# 444203) was added to reduce samples. Reduced and non-reduced samples were resolved on Novex NuPAGE gels (4–12% bis-tris) (Invitrogen, Cat# NP0323BOX) in MES buffer (Invitrogen, Cat# NP0002), transferred onto PVDF membrane (Invitrogen, Cat# LC2005) before blocking in 5% milk in PBST for 1 h. Membranes were incubated with Hu IgG (Fc/γ Chain)-PEROX (The Binding Site, Cat# AP004) to detect HC and Hu Kappa (AFF)-PEROX (The Binding Site, Cat# AP015) to detect LC, both diluted at 1:3000 in 5% milk in PBST. Membranes were developed using ECL reagent (Amersham, Cat# RPN2105V1 + 2).

Abbreviations

CHO	Chinese hamster ovary
CSR	class switch recombination
CQA	critical quality attributes
GS	glutamine synthetase
HC	heavy chain
HCCD	heavy chain constant domain
IgG	immunoglobulin
IVC	integral of viable cells (10 ⁹ cell hr/L)
LC	light chain
mAbs	monoclonal antibodies
qP	cell productivity (pg/(cell day))
VCN	viable cell number (x10 ⁶ /mL)
3'SS	3' splice site

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Author contributions

All authors provided input into the research and manuscript. E.K., C.H., D.H., S.D., and S.G. conceptualized and designed the experiments; E.K. and T.S. performed experiments and analyzed the data and E.K., T.S., D.H., S.D., and S.G. wrote and reviewed the manuscript.

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